State Space Analysis Findings: Gamma and Alpha Bands

Project: Characterizing the Neural State Space Dynamics of Long-Term Rajyoga Meditation **Authors:** Muralidhar Rao M, Prateek Kumar, Shreeansh Hota **Date:** April 2026 **Dataset:** Biswas et al. — 30 Rajyoga meditator-control pairs, 64-channel EEG **Conditions:** EO1 (eyes-open baseline, 5 min) vs M1 (eyes-open meditation, 15 min)

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1. Analysis Pipeline

Raw segmented EEG trials were bandpass filtered (4th-order Butterworth, zero-phase), downsampled, and concatenated across good trials to form continuous electrode-time trajectories. Trials were concatenated rather than averaged because spontaneous oscillations are not phase-locked to trial onset and would cancel under averaging. Bad trials were rejected using a three-layer pipeline inherited from the Biswas et al. preprocessing code: (1) eye position excursions, (2) RMS amplitude thresholding (1.5-35 uV), and (3) PSD outlier detection (mean + 6 SD). Minimum thresholds: 30 good trials and 10 good electrodes per subject-protocol.

The analysis framework had three tiers:

Tier 1 — Signal Subspace Extraction: PCA was applied to each subject's electrode-time matrix. Horn's Parallel Analysis (100 permutations, 95th percentile threshold) determined the individual embedding dimension k. The null distribution is generated by **Phase Randomization** of each electrode's FFT (with conjugate symmetry preserved so the IFFT is real), producing colored-noise surrogates that match the bandpassed power spectrum of the data. This replaces the earlier time-shuffling (`randperm`) approach, which destroyed autocorrelation and produced white-noise surrogates that artificially lowered the threshold and inflated k. A global standardized dimension K_std was set as the median k of Controls at EO1 baseline. All subsequent linear metrics were computed in this common K_std-dimensional space to enable fair cross-subject comparison of volume.

Tier 2 — Dynamical Metrics: Four markers were computed in the K_std-dimensional PCA space:

- **Trajectory Velocity:** Mean Euclidean step size between consecutive *within-trial* time points (proxy for state volatility). Step sizes spanning the boundary between concatenated trials are masked out before averaging — these would otherwise inject artificial "teleportation" spikes from the discontinuity between the last sample of trial N and the first sample of trial N+1.
- **State Space Volume:** Log hyper-ellipsoid volume from eigenvalues of the projected covariance (quantifies dynamic range). Eigenvalues are clamped at a floor of 1e-6 (rather than MATLAB's `eps` ≈ 2.2×10^{-16}), so subjects with locally low-rank covariance are penalized gracefully rather than catastrophically dragged into a -36 hole by `log(eps)`.
- **Excursion Radius:** Mean distance of trajectory points from the geometric centroid (measures spatial spread).
- **Participation Ratio (PR):** PR = $(\Sigma\lambda)^2 / \Sigma\lambda^2$, computed from the full PCA eigenvalue spectrum. PR is an amplitude-independent measure of effective linear dimensionality: PR = 1 when all variance is in one

component, PR = N when variance is distributed uniformly across N components. Unlike volume/velocity/radius, PR is a pure ratio and is not inflated by total signal power.

Tier 3 — Intrinsic Complexity: The Two-Nearest Neighbors (TNN) estimator (Facco et al., 2017) was applied to the full-dimensional data (not k-truncated) to estimate nonlinear intrinsic dimension d. If full dimensionality caused memory constraints, PCA retaining 99% of variance was applied first. Data was subsampled to 10,000 points via uniform random sampling when trajectory length exceeded this threshold.

Statistics: All between-group comparisons used paired Wilcoxon signed-rank tests (matched meditator-control pairs). Within-group EO1 vs M1 comparisons also used paired Wilcoxon signed-rank. To disentangle structural expansion from amplitude inflation in the volume metric, an **ANCOVA** (multiple linear regression) was additionally fit per protocol: `LogVolume ~ Group + TotalVariance`, with Group as a categorical predictor (Med = 1, Con = 0) and TotalVariance (sum of all PCA eigenvalues, a proxy for total signal power) as a continuous covariate. The p-value on the Group coefficient tests whether the volume difference between meditators and controls survives controlling for power. Significance thresholds: * $p<0.05$, ** $p<0.01$, *** $p<0.005$.

Important note on metric interpretation: Velocity, volume, and radius are amplitude-dependent — they scale with signal power. A signal that is 2x louder will produce approximately 2x velocity and 2x radius regardless of dynamical complexity. The embedding dimension k, Participation Ratio (PR), and TNN intrinsic dimension d are amplitude-independent — they measure the number of independent degrees of freedom irrespective of signal magnitude. This distinction is critical for interpreting group differences: k, PR, d, and the ANCOVA-adjusted volume all test the dimensionality / structural hypothesis directly, while raw velocity/volume/radius confound dynamical structure with power.

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2. Key Parameters

Parameter	Gamma Analysis	Alpha Analysis
Frequency band	30-80 Hz	8-13 Hz
Filter	4th-order Butterworth bandpass, zero-phase	Same
Downsampled Fs	250 Hz	50 Hz
Trial handling	Concatenation (within-trial velocity only)	Same
Parallel analysis surrogate	Phase Randomization (colored noise)	Same
Parallel analysis permutations	100	100
Eigenvalue clamp floor	1e-6	1e-6
K_std	5	3
TNN subsampling	10,000 points (uniform random)	Same
Valid pairs (EO1)	29/30	29/30
Valid pairs (M1)	29/30	29/30
Missing subjects	053DR EO1 (no data), 019CKa M1 (5 trials)	Same

Both bands have now been processed under the upgraded pipeline (Phase Randomization surrogates, within-trial velocity, 1e-6 eigenvalue clamp, Participation Ratio, ANCOVA on volume).

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3. Gamma-Band Results (30-80 Hz, K_std = 5)

3.1 Between-Group: Meditators vs Controls

Metric	Condition	Meditators (mean +/- SEM)	Controls (mean +/- SEM)	p-value	Direction
k	EO1	6.41 +/- 0.50	5.23 +/- 0.47	0.135	N.S.
k	M1	6.52 +/- 0.47	5.93 +/- 0.44	0.472	N.S.
Velocity	EO1	25.27 +/- 1.96	20.41 +/- 1.78	0.074	Med > Con (trend)
Velocity	**M1**	**28.85 +/- 2.11**	**20.89 +/- 1.54**	**0.007 ★★**	**Med > Con**
Log Volume	EO1	12.20 +/- 0.38	10.93 +/- 0.33	0.020 ★	Med > Con
Log Volume	**M1**	**13.18 +/- 0.38**	**11.59 +/- 0.38**	**0.013 ★**	**Med > Con**
Radius	EO1	21.36 +/- 1.62	17.23 +/- 1.50	0.048 ★	Med > Con
Radius	**M1**	**24.26 +/- 1.77**	**17.66 +/- 1.28**	**0.009 ★★**	**Med > Con**
PR	EO1	5.81 +/- 0.41	5.27 +/- 0.42	0.552	N.S.
PR	M1	6.02 +/- 0.39	5.04 +/- 0.39	0.065	Med > Con (trend)
TNN d	EO1	18.39 +/- 0.39	18.61 +/- 0.53	0.871	N.S.
TNN d	M1	17.77 +/- 0.35	17.94 +/- 0.43	0.705	N.S.

3.2 Within-Group: EO1 vs M1

Metric	Group	EO1 (mean +/- SEM)	M1 (mean +/- SEM)	p-value	Direction
k	Meditators	6.41 +/- 0.51	6.52 +/- 0.48	0.673	No change
k	Controls	5.23 +/- 0.47	5.93 +/- 0.43	0.084	Trending up
Velocity	Meditators	25.27 +/- 2.00	28.85 +/- 2.15	0.106	No change
Velocity	Controls	20.41 +/- 1.75	20.89 +/- 1.51	0.558	No change
Log Volume	**Meditators**	**12.20 +/- 0.38**	**13.18 +/- 0.38**	**0.006 ★★**	**Increases**
Log Volume	Controls	10.93 +/- 0.32	11.59 +/- 0.38	0.032 ★	Increases

Metric	Group	EO1 (mean +/- SEM)	M1 (mean +/- SEM)	p-value	Direction
Radius	Meditators	21.36 +/- 1.65	24.26 +/- 1.80	0.122	No change
Radius	Controls	17.23 +/- 1.48	17.66 +/- 1.26	0.530	No change
PR	Meditators	5.81 +/- 0.42	6.02 +/- 0.40	0.873	No change
PR	Controls	5.27 +/- 0.41	5.04 +/- 0.38	0.910	No change
TNN d	Meditators	18.39 +/- 0.40	17.77 +/- 0.35	0.111	No change
TNN d	Controls	18.61 +/- 0.52	17.94 +/- 0.42	0.047 ★	Decreases

3.3 ANCOVA: Disentangling Volume from Power

To test whether the LogVolume group difference is driven by amplitude alone, we fit `LogVolume ~ Group + TotalVariance` per protocol. TotalVariance (sum of all PCA eigenvalues) serves as a proxy for total gamma power.

Protocol	Predictor	Coefficient	SE	p-value
EO1	Group (Med vs Con)	**+0.646**	0.283	**0.026 ★**
EO1	TotalVariance	+0.00256	0.00023	<0.0001 ★★★
EO1	(R^2 = 0.732)			
M1	**Group (Med vs Con)**	**+1.505**	0.435	**0.0011 ★★★**
M1	TotalVariance	+0.00065	0.00012	<0.0001 ★★★
M1	(R^2 = 0.445)			

The TotalVariance coefficient is highly significant in both protocols, confirming that gamma power is a strong predictor of state space volume. **Critically, the Group coefficient remains significant after controlling for TotalVariance**: meditators' volumes are larger than controls' even at matched power levels. The Group effect roughly *doubles* from EO1 (+0.65) to M1 (+1.50), indicating that the structural expansion is amplified during meditation rather than being a fixed trait offset. Power explains less of the variance at M1 (R^2 drops from 0.73 to 0.45), leaving more room for genuine structural Group differences to emerge.

3.4 Gamma Summary

The amplitude-dependent metrics (velocity, volume, radius) show robust group differences at M1 (all $p \leq 0.013$), with meditators exhibiting larger, faster, and more expansive trajectories. These metrics are confounded with gamma power, and Biswas et al. already established that meditators have higher gamma power. However, the ANCOVA result demonstrates that **the LogVolume group difference is not fully reducible to power** — Group remains a significant predictor ($p = 0.0011$ at M1) even after adjusting for TotalVariance. Consistent with this, the amplitude-independent **Participation Ratio shows a Med > Con trend at M1** ($p = 0.065$), suggesting meditators distribute variance across more effective dimensions during meditation. The nonlinear intrinsic dimension (TNN d) shows no group difference ($p > 0.7$), indicating the *manifold topology* is unchanged even as the *linear effective dimensionality* trends apart.

Within-group, the meditation state primarily increases state space volume in both groups (Med p = 0.006, Con p = 0.032). Other metrics show only modest within-group changes after correcting the velocity calculation.

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4. Alpha-Band Results (8-13 Hz, K_std = 3)

4.1 Between-Group: Meditators vs Controls

Metric	Condition	Meditators (mean +/- SEM)	Controls (mean +/- SEM)	p-value	Direction
k	EO1	3.52 +/- 0.15	3.27 +/- 0.10	0.168	N.S.
k	M1	3.83 +/- 0.22	3.47 +/- 0.18	0.231	N.S.
Velocity	EO1	20.44 +/- 2.24	19.64 +/- 1.68	0.991	N.S.
Velocity	M1	21.49 +/- 2.32	22.63 +/- 1.96	0.552	N.S.
Log Volume	EO1	7.98 +/- 0.28	8.14 +/- 0.25	0.642	N.S.
Log Volume	M1	8.32 +/- 0.27	8.65 +/- 0.29	0.430	N.S.
Radius	EO1	16.85 +/- 1.88	16.08 +/- 1.37	0.974	N.S.
Radius	M1	17.74 +/- 1.92	18.69 +/- 1.64	0.524	N.S.
PR	EO1	2.68 +/- 0.14	2.58 +/- 0.13	0.705	N.S.
PR	M1	2.61 +/- 0.11	2.49 +/- 0.15	0.405	N.S.
TNN d	EO1	10.78 +/- 0.24	10.25 +/- 0.23	0.157	N.S.
TNN d	**M1**	**12.18 +/- 0.32**	**10.86 +/- 0.35**	**0.037 ★**	**Med > Con**

4.2 Within-Group: EO1 vs M1

Metric	Group	EO1 (mean +/- SEM)	M1 (mean +/- SEM)	p-value	Direction
k	Meditators	3.52 +/- 0.16	3.83 +/- 0.23	0.108	No change
k	Controls	3.27 +/- 0.10	3.47 +/- 0.18	0.341	No change
Velocity	Meditators	20.44 +/- 2.28	21.49 +/- 2.36	0.011 ★	Increases
Velocity	**Controls**	**19.64 +/- 1.65**	**22.63 +/- 1.93**	**0.0002 ★★★**	**Increases**
Log Volume	Meditators	7.98 +/- 0.29	8.32 +/- 0.28	0.024 ★	Increases

Metric	Group	EO1 (mean +/- SEM)	M1 (mean +/- SEM)	p-value	Direction
Log Volume	**Controls**	**8.14 +/- 0.25**	**8.65 +/- 0.29**	**0.0008 ★★★**	**Increases**
Radius	Meditators	16.85 +/- 1.91	17.74 +/- 1.96	0.008 ★★	Increases
Radius	**Controls**	**16.08 +/- 1.35**	**18.69 +/- 1.61**	**0.0001 ★★★**	**Increases**
PR	Meditators	2.68 +/- 0.15	2.61 +/- 0.11	0.439	No change
PR	Controls	2.58 +/- 0.13	2.49 +/- 0.15	0.349	No change
TNN d	**Meditators**	**10.78 +/- 0.25**	**12.18 +/- 0.32**	**<0.0001 ★★★**	**Increases**
TNN d	Controls	10.25 +/- 0.23	10.86 +/- 0.34	0.014 ★	Increases

4.3 ANCOVA: LogVolume ~ Group + TotalVariance (Alpha)

Protocol	Predictor	Coefficient	SE	p-value
EO1	Group (Med vs Con)	−0.314	0.194	0.111
EO1	TotalVariance	+0.00163	0.00013	<0.0001 ★★★
EO1	(R^2 = 0.744)			
M1	Group (Med vs Con)	−0.107	0.320	0.739
M1	TotalVariance	+0.00049	0.00008	<0.0001 ★★★
M1	(R^2 = 0.403)			

In sharp contrast to the gamma ANCOVA, the alpha Group coefficient is **negative** in both protocols (Meditators have *smaller* power-adjusted volume than Controls) and **non-significant** in both. TotalVariance is, as expected, a strong predictor of LogVolume. The negative direction is consistent with the raw means (Med < Con on volume in both protocols, though not significantly), but neither raw nor power-adjusted differences reach significance. The structural-expansion finding seen in gamma does not replicate in alpha.

4.4 Alpha Summary

Between-group comparisons show no differences on any amplitude-dependent metric (all $p > 0.4$), no PR difference ($p > 0.4$), and no power-adjusted volume difference (ANCOVA Group $p > 0.1$ in both protocols). This is consistent with the Biswas et al. finding that the meditator-control power difference is concentrated in the gamma band, not alpha. The one significant between-group result is TNN intrinsic dimension at M1 ($p = 0.037$), but in the opposite direction from the hypothesis: meditators show higher, not lower, nonlinear complexity during meditation. Notably, the **linear** effective dimensionality (PR) is at chance — so the alpha group difference is purely a *nonlinear* phenomenon, not a redistribution of linear PCA variance.

Within-group, the alpha band shows a striking universal state effect: velocity, volume, radius, and TNN d all increase from EO1 to M1 in both groups (PR and k do not change). Notably, controls show stronger significance than meditators on the amplitude-dependent metrics (velocity p=0.0002 vs 0.011; volume p=0.0008 vs 0.024; radius p=0.0001 vs 0.008). The meditation condition expands the alpha state space in everyone, and controls respond at least as strongly as meditators on these measures.

The key differentiator is TNN d: meditators show a highly significant increase from 10.78 to 12.18 ($p < 0.0001$), while controls increase more modestly from 10.25 to 10.86 ($p = 0.014$). Meditation increases the nonlinear complexity of alpha dynamics in both groups, but experienced meditators undergo a substantially larger increase.

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5. Comparison: Gamma vs Alpha

5.1 Between-Group Differences at M1

Metric	Gamma Direction	Gamma p	Alpha Direction	Alpha p
k	N.S.	0.472	N.S.	0.231
Velocity	Med > Con	0.007 ★★	N.S.	0.552
Log Volume (raw)	Med > Con	0.013 ★	N.S.	0.430
Log Volume (ANCOVA, power-adjusted)	**Med > Con**	**0.0011 ★★★**	N.S. (Med < Con, n.s.)	0.739
Radius	Med > Con	0.009 ★★	N.S.	0.524
PR	Med > Con (trend)	0.065	N.S.	0.405
TNN d	N.S.	0.705	Med > Con	0.037 ★

Gamma separates meditators from controls on amplitude-dependent metrics (velocity, volume, radius) and shows a trend toward elevated effective dimensionality (PR). The ANCOVA confirms the volume difference is *not* purely driven by gamma power — Group remains highly significant ($p = 0.0011$) after adjusting for TotalVariance. The nonlinear intrinsic dimensionality (TNN d) is unchanged, so the manifold topology is preserved while its linear extent expands.

Alpha shows no amplitude-dependent group differences (consistent with no alpha power difference between groups), no Participation Ratio difference (PR is at chance), and no power-adjusted volume difference (ANCOVA Group coefficient is negative and non-significant). The only group difference is on TNN d at M1, in the direction opposite to the original hypothesis. The alpha group difference is therefore *purely nonlinear* — it does not show up in any linear measure (raw means, ANCOVA, PR), only in the nonlinear intrinsic-dimension estimator.

5.2 Within-Group EO1 to M1 Changes

Metric	Gamma Med p	Gamma Con p	Alpha Med p	Alpha Con p
k	0.673	0.084	0.108	0.341
Velocity	0.106	0.558	0.011 ★	0.0002 ★★★
Log Volume	0.006 ★★	0.032 ★	0.024 ★	0.0008 ★★★
Radius	0.122	0.530	0.008 ★★	0.0001 ★★★
PR	0.873	0.910	0.439	0.349
TNN d	0.111	0.047 ★	<0.0001 ★★★	0.014 ★

The within-group pattern is fundamentally different between bands:

In gamma, the meditation state effect is concentrated in volume — meditators show a robust increase (p = 0.006) and controls a smaller but significant increase (p = 0.032). Velocity and radius no longer reach significance for either group once concatenation-boundary spikes are removed. PR is essentially flat within-group, suggesting the volume increase from EO1 to M1 is driven more by amplitude growth than by spreading variance into additional dimensions.

In alpha, the meditation state effect is universal but stronger in controls for amplitude metrics. Both groups show highly significant increases across all metrics, but controls reach stronger significance (velocity p=0.0002, radius p=0.0001) than meditators (velocity p=0.011, radius p=0.008). The exception is TNN d, where meditators show the stronger effect (p<0.0001 vs p=0.011).

This dissociation suggests that the meditation state globally expands the alpha state space in everyone (possibly reflecting increased alpha synchronization during sustained eyes-open rest), and controls may even show larger amplitude-level changes. However, the structural reorganization of the alpha manifold — as captured by nonlinear intrinsic dimensionality — is more pronounced in experienced meditators.

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6. Hypothesis Evaluation

The "Shape of Stillness" hypothesis predicted that meditation would induce a collapse of neural degrees of freedom: lower trajectory velocity, smaller state space volume, reduced excursion radius, and lower intrinsic dimensionality in meditators compared to controls, particularly during M1.

6.1 Predictions vs Observations

#	Prediction	Gamma Result	Alpha Result	Overall Verdict
1	Med velocity < Con at M1	Med significantly higher (p=0.007)	No difference (p=0.552)	**Rejected**
2	Med volume < Con at M1	Med significantly higher (raw p=0.013; ANCOVA Group p=0.0011)	No difference (p=0.430)	**Rejected**
3	Med radius < Con at M1	Med significantly higher (p=0.009)	No difference (p=0.524)	**Rejected**

#	Prediction	Gamma Result	Alpha Result	Overall Verdict
4	Med PR < Con at M1	Med trending higher (p=0.065)	No difference (p=0.405)	**Rejected**
5	Med TNN d < Con at M1	No difference (p=0.705)	Med significantly higher (p=0.037)	**Rejected**
6	Velocity drops EO1 to M1 in Med	No significant change (p=0.106)	Increases (p=0.011)	**Rejected**
7	TNN d drops EO1 to M1 in Med	No significant change (p=0.111)	Increases (p<0.0001)	**Rejected**
8	Lower std(TNN d) in Med at M1	Similar variance	Med std=1.71, Con std=1.86	**Weak/inconclusive**

The "Shape of Stillness" hypothesis is not supported in either the gamma or alpha band. No metric, in either band, showed the predicted reduction in meditators. The amplitude-dependent metrics either favor meditators having larger dynamics (gamma) or show no group difference (alpha). The amplitude-independent dimensionality metrics either show no difference (gamma k and TNN d; alpha k) or show meditators having *higher* complexity (alpha TNN d at M1; gamma PR trend at M1). Critically, the ANCOVA-adjusted volume in gamma — which controls for the obvious power confound — also shows meditators expanded, not collapsed.

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7. Interpretation

7.1 Gamma band: Amplitude *and* structural expansion, not dimensional collapse

The gamma results are consistent with the established finding that experienced meditators exhibit elevated broadband gamma power (Biswas et al., 2024). The state space analysis adds that this power difference manifests as faster, more expansive neural trajectories. Crucially, the **ANCOVA result rules out the interpretation that the volume expansion is purely a power artifact**: even after regressing out TotalVariance, Group remains a highly significant predictor of LogVolume ($p = 0.0011$ at M1, $p = 0.026$ at EO1), and the Group effect more than doubles from EO1 to M1. The Participation Ratio, which is mathematically immune to amplitude scaling, shows the same pattern in trend form (Med > Con at M1, $p = 0.065$). Together, these indicate that meditators distribute gamma variance across slightly more effective dimensions during meditation — a *structural* expansion on top of the well-established power expansion. The nonlinear manifold topology (TNN d ~ 18 in both groups) remains unchanged, so this is a redistribution of variance within the existing manifold rather than a recruitment of new nonlinear degrees of freedom.

7.2 Alpha band: Nonlinear complexity increase, not collapse — and purely nonlinear

The alpha results challenge the hypothesis most directly. The one amplitude-independent metric that does separate the groups — TNN intrinsic dimension at M1 — goes in the opposite direction: meditators show higher nonlinear complexity ($d = 12.18$) than controls ($d = 10.86$) during meditation. Furthermore, within meditators, TNN d increases from 10.78 at baseline to 12.18 during meditation ($p < 0.0001$), while controls show a smaller increase (10.25 to 10.86, $p = 0.014$).

The upgraded pipeline sharpens this picture. The new alpha **ANCOVA shows no power-adjusted group difference** in volume (Group p = 0.74 at M1, with the coefficient even slightly negative — Med below Con). The new alpha **Participation Ratio is also at chance** in both protocols (p > 0.4) and changes negligibly within-group between EO1 and M1 (p > 0.3). So the alpha group difference is *purely a nonlinear phenomenon* — it is invisible to every linear measure (raw means, ANCOVA-adjusted volume, PR) and appears only in the nonlinear intrinsic-dimension estimator.

This suggests that meditation does not simplify the alpha manifold but instead enriches it in a way that cannot be captured by linear PCA-based measures. Experienced practitioners may recruit more independent alpha generators or engage in more complex patterns of alpha synchronization/desynchronization during focused attention — patterns that fold the trajectory through additional nonlinear directions without changing how variance is distributed across linear PCA components.

7.3 Band-specific dissociation

The most notable finding is the dissociation between bands:

- **Gamma:** Group differences are largely amplitude-driven (power), but a *residual linear-structural component* remains after power adjustment (ANCOVA Group p = 0.0011 at M1) and is mirrored by a PR trend (p = 0.065). Nonlinear topology (TNN d) is unchanged. The meditation state preferentially amplifies gamma in meditators, *and* slightly enriches its linear effective dimensionality.
- **Alpha:** No linear group differences exist on any measure — raw amplitude metrics, power-adjusted volume (ANCOVA), and PR are all at chance. The sole group difference is *nonlinear* (TNN d), with meditators showing increased intrinsic dimension. The meditation state activates alpha dynamics universally on linear measures, with controls showing equal or stronger amplitude changes.

This is a crisp band-by-mechanism dissociation. Gamma differences are linear-structural (variance gets redistributed across PCA components, plus more total power). Alpha differences are nonlinear-structural (the manifold develops more independent directions in a way only TNN can detect, with no linear PCA fingerprint at all). Neither band shows the predicted dimensional collapse.

7.4 Alternative framing

Rather than a "collapse of degrees of freedom," long-term meditation practice may produce an "expansion of controlled complexity" — the ability to sustain more independent neural modes simultaneously during focused attention, particularly in the alpha band (TNN d), while also generating stronger gamma-band activity that itself is modestly more dimensionally distributed (gamma PR trend, ANCOVA Group effect on volume). The meditator's brain during meditation is not simpler than a control's; it is more richly structured both in the linear sense (more equally distributed PCA variance in gamma) and in the nonlinear sense (higher TNN intrinsic dimension in alpha).

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8. Pair-Level Consistency Analysis

Group-level statistics can be driven by a few outlier pairs. To verify that the significant between-group results reflect distributed effects, we examined the direction of the meditator-control difference for each of the 29 valid matched pairs at both EO1 and M1, along with a two-tailed exact binomial test against a 50/50 null (ties excluded from the test).

8.1 Gamma Band — Pair Consistency at EO1

Metric	Med > Con	Med < Con	Med = Con	% Med > Con	Binomial p
k	17	10	2	58.6%	0.248
Velocity	19	10	0	65.5%	0.136
Log Volume	19	10	0	65.5%	0.136
Radius	20	9	0	69.0%	0.061
PR	15	14	0	51.7%	1.000
TNN d	16	13	0	55.2%	0.711

At baseline (EO1), the amplitude metrics already lean Med > Con (65–69% of pairs), with Radius approaching significance (p = 0.061). PR and TNN d are at chance level, consistent with the group-level result that no dimensionality difference exists at baseline.

8.2 Gamma Band — Pair Consistency at M1

Metric	Med > Con	Med < Con	Med = Con	% Med > Con	Binomial p
k	15	10	4	51.7%	0.424
Velocity	**22**	**7**	**0**	**75.9%**	**0.008**
Log Volume	19	10	0	65.5%	0.136
Radius	**22**	**7**	**0**	**75.9%**	**0.008**
PR	18	11	0	62.1%	0.265
TNN d	12	17	0	41.4%	0.458

During meditation, the gamma Velocity and Radius results are highly consistent at the pair level: 22 out of 29 matched pairs (75.9%) show the meditator having higher values than their matched control, and the binomial test against 50/50 yields p = 0.008 for both. These effects are clearly distributed across the sample, not driven by a few outlier pairs. Log Volume shows the same direction (19/29, 65.5%) but does not cross the per-pair binomial threshold despite being significant at the group level — indicating the volume effect is more about *magnitude per pair* than *consistency across pairs*. The PR trend from §3.1 (group p = 0.065) is mirrored at the pair level (18/29 = 62.1%, binomial p = 0.265), suggesting the structural-expansion signal is real but smaller than the amplitude effects. The k and TNN d counts are at chance, confirming no directional consistency on dimensionality metrics in gamma.

8.3 Alpha Band — Pair Consistency at EO1

Metric	Med > Con	Med < Con	Med = Con	% Med > Con	Binomial p
k	11	5	13	37.9%	0.210
Velocity	16	13	0	55.2%	0.711

Metric	Med > Con	Med < Con	Med = Con	% Med > Con	Binomial p
Log Volume	15	14	0	51.7%	1.000
Radius	15	14	0	51.7%	1.000
PR	17	12	0	58.6%	0.458
TNN d	18	11	0	62.1%	0.265

At baseline, all alpha metrics are essentially at chance, including TNN d which only trends Med > Con (62%, p = 0.27). This is consistent with no group difference at EO1 in any group-level alpha test.

8.4 Alpha Band — Pair Consistency at M1

Metric	Med > Con	Med < Con	Med = Con	% Med > Con	Binomial p
k	12	8	9	41.4%	0.503
Velocity	16	13	0	55.2%	0.711
Log Volume	15	14	0	51.7%	1.000
Radius	15	14	0	51.7%	1.000
PR	18	11	0	62.1%	0.265
TNN d	**21**	**8**	**0**	**72.4%**	**0.024**

The alpha TNN d result holds at the pair level under the upgraded pipeline: 21 out of 29 pairs show the meditator having higher intrinsic dimension during meditation than their matched control (binomial p = 0.024) — identical to the legacy pipeline counts. The effect sizes are modest but consistent (most differences in the range of +0.5 to +5.0), with the 8 pairs that go in the opposite direction showing small negative differences (−0.3 to −4.1). Every other alpha metric — including the new PR — is at chance, confirming that the meditator-control alpha difference manifests only in the nonlinear estimator.

8.5 Summary

The primary findings — elevated gamma amplitude metrics in meditators at M1 and elevated alpha TNN d at M1 — are distributed effects present in approximately 72–76% of matched pairs, not artifacts of a few extreme subjects. Conversely, the null results (gamma TNN d, gamma k, all alpha amplitude metrics, alpha PR) are confirmed as true nulls at the pair level, with splits indistinguishable from chance. The new gamma PR trend (18/29 = 62%) goes in the predicted direction but does not reach pair-level significance, mirroring the group-level trend.

Finding	Pairs supporting	Pairs opposing	Pair-level binomial p
Gamma velocity Med > Con at M1	22/29 (76%)	7/29 (24%)	0.008
Gamma radius Med > Con at M1	22/29 (76%)	7/29 (24%)	0.008
Gamma volume Med > Con at M1	19/29 (66%)	10/29 (34%)	0.136
Gamma PR Med > Con at M1 (trend)	18/29 (62%)	11/29 (38%)	0.265

Finding	Pairs supporting	Pairs opposing	Pair-level binomial p
Alpha TNN d Med > Con at M1	21/29 (72%)	8/29 (28%)	0.024
Gamma TNN d (null)	12/29 (41%)	17/29 (59%)	0.458
Gamma k (null)	15/25 (52%, 4 ties)	10/25 (40%)	0.424
Alpha volume (null)	15/29 (52%)	14/29 (48%)	1.000
Alpha PR (null)	18/29 (62%)	11/29 (38%)	0.265

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9. Limitations

- Parallel analysis used 100 permutations (reduced from 1000 for computational feasibility); k estimates may have slightly higher variance. Surrogates are now generated by Phase Randomization (preserves the bandpassed power spectrum), which is more conservative than the prior time-shuffle approach and reduced K_std in gamma from 6 to 5.
- TNN subsampling to 10,000 points via uniform random sampling introduces stochasticity; results may vary slightly across runs.
- Velocity, volume, and radius are confounded with signal amplitude and cannot distinguish power differences from dynamical structure differences. The ANCOVA on volume (§3.3) and the Participation Ratio partially address this; k and TNN d provide fully amplitude-independent tests of the dimensionality hypothesis.
- Two subject-protocol combinations were excluded in both analyses (019CKa M1: insufficient good trials; 053DR EO1: missing data).
- The alpha TNN d group difference ($p = 0.037$) would not survive Bonferroni correction across the between-group tests per band. It should be interpreted as exploratory.
- The gamma PR trend at M1 ($p = 0.065$) likewise does not cross $\alpha = 0.05$ and should be treated as suggestive.
- The analysis does not address broadband or cross-frequency dynamics, which may be relevant to the hypothesis.